

// PostgreSQL

// CONNECT

```
docker compose -f infra/compose.yaml \
  up -d postgres
docker compose ... \
  exec postgres psql -U orders -d orders
# prompt: orders=#
```

image `postgres:18`

db/user `orders / orders`

jdbc `jdbc:postgresql://postgres:5432/orders`

// PSQL META-COMMANDS

```
\l          list databases
\d          list schemas
\dt         list tables
\d applicants describe table
\du         list roles
\timing on  time every query
\q         quit
```

// CREATE TABLE

```
CREATE TABLE applicants (
  id          bigint GENERATED ALWAYS AS IDENTITY
              PRIMARY KEY,
  national_id varchar NOT NULL UNIQUE,
  full_name   varchar NOT NULL,
  country     varchar NOT NULL,
  status      varchar NOT NULL DEFAULT 'PENDING'
              CHECK (status IN ('PENDING','CLEARED','BLOCKED')),
  attributes  jsonb,
  created_at  timestampz NOT NULL DEFAULT now()
);
```

// INSERT · RETURNING · UPSERT

```
INSERT INTO applicants (national_id,full_name,country)
VALUES ('NID-1001','Ada Lovelace','GB')
RETURNING id;
```

```
INSERT INTO applicants (national_id,full_name,country)
VALUES ('NID-1001','Ada L.','GB')
ON CONFLICT (national_id) DO UPDATE
  SET full_name = EXCLUDED.full_name;
```

// FOREIGN KEY

```
CREATE TABLE screenings (
  id          bigint GENERATED ALWAYS AS IDENTITY PK,
  applicant_id bigint NOT NULL
              REFERENCES applicants(id),
  result      varchar NOT NULL,
  screened_at timestampz DEFAULT now()
);
```

// JOIN · AGGREGATE · WINDOW

```
SELECT a.country, count(*)
FROM screenings s
JOIN applicants a ON a.id = s.applicant_id
GROUP BY a.country;
```

```
SELECT full_name, country,
       rank() OVER (PARTITION BY country
                    ORDER BY created_at)
FROM applicants;
```

// JSONB

```
UPDATE applicants SET attributes =
  '{"risk":"low","tags":["vip"]}'
WHERE national_id = 'NID-1001';

-- → extract as text
SELECT attributes→'risk' FROM applicants;
-- @> contains
SELECT * FROM applicants
WHERE attributes @> '{"risk":"low"}';
```

// EXPLAIN → INDEX → EXPLAIN

```
EXPLAIN SELECT * FROM applicants
  WHERE national_id = 'NID-1001';  # Seq Scan
```

```
CREATE INDEX ON applicants (national_id);
```

```
EXPLAIN ...  # Index Scan
```

// TRANSACTIONS

```
BEGIN;
UPDATE applicants SET status='BLOCKED'
  WHERE country = 'XX';
ROLLBACK;  # or COMMIT
```

// ROLES / GRANTS

```
CREATE ROLE kyc_readonly NOLOGIN;
GRANT CONNECT ON DATABASE orders TO kyc_readonly;
GRANT USAGE ON SCHEMA public TO kyc_readonly;
GRANT SELECT ON ALL TABLES IN SCHEMA public
  TO kyc_readonly;
```

// MAINTENANCE & TROUBLESHOOTING

```
VACUUM (VERBOSE, ANALYZE) applicants;
SELECT * FROM pg_stat_activity;
SELECT pg_cancel_backend(<pid>);
```

// BACKUP

```
docker compose ... exec -T postgres \
  pg_dump -U orders orders > kyc-backup.sql
```